Muhammad Moiz Khalid

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EDUCATION

NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES (NUCES - FAST) | Islamabad, Pakistan
BS in Data Science | Aug 2023 – May 2027 | CGPA: 3.82/4.0
Major: *Data Science*

- *Gold Medal – Fall'24 & Spring'25*
- *Silver Medal – Fall'23*
- *Dean's List: Fall'23, Spring'24, Fall'24, Spring'25, Fall'25*
- *Relevant Coursework: Artificial Intelligence, Intro to Data Science, Database Systems, Advanced Statistics, Data Mining, Data Warehousing & Business Intelligence, Deep Learning, Design & Analysis of Algorithms, NLP*

Bahria College Islamabad | Islamabad, Pakistan
Cambridge O-Level | Cambridge A-Level | Graduated June 2023 | GPA: 4.0/4.0 (O-Level & A-Level)

- *100% Merit Scholarship for Top O-Level Result (2021)*
- *A-Level Grades: Math: A, Physics: A, Chemistry: A, Computer Science: A*
- *O-Level Grades: 8 A*s, 1 A (incl. Math, Computer Science, Add Math, Physics, Chemistry)*

WORK EXPERIENCE

Undergraduate Researcher, *NUCES - FAST*
Computer Vision & Deep Learning | Jan 2025 - Present

- Chosen as one of five students for a competitive research role based on top performance in a relevant course.
- Building a strong foundation in computer vision and deep learning under faculty mentorship.
- Preparing for active research by reviewing recent work, studying model architectures, and exploring key datasets.

Data Annotator, *Turing* | Jun 2025 – Aug 2025

- Contributed to Anthropic's agentic AI LLM project by generating and refining high-quality training data using proprietary tools.
- Summarized complex content into structured, accurate, and well-reasoned outputs.
- Verified facts, handled reasoning-intensive prompts, and ensured all data met strict quality standards.
- Developed strong analytical skills and gained hands-on experience with high-precision data workflows for advanced AI systems.

Teaching Assistant / Lab Demonstrator, *NUCES - FAST*
OOP Lab (Fall'24) | PF Lab (Spring'25) | IDS Lab (Fall'25) | PF Theory (Fall'25)

- As a Programming Fundamentals TA, I helped students understand core programming concepts, created and graded quizzes and assignments, supervised semester projects, and supported them throughout the course.
- In labs, guided 60+ undergraduates through one-on-one and group support, co-designed lab exercises, delivered hands-on demonstrations, and provided debugging assistance in C++, OOP, and Python.

PROJECTS

Case Study: Energy Consumption in London *(Python)*

- Predicted energy usage patterns using demographic, weather, and temporal data from 3.5M+ smart meter records.
- Cleaned data and engineered features to condense the dataset to 15,000 representative entries for efficient modeling.
- Evaluated multiple models: Linear Regression, Ridge, LASSO, Elastic Net, Decision Tree, Random Forest, and Gradient Boosting Regressor, with tree-based methods significantly outperforming others.

Near-Real-Time Data Warehouse for Walmart using HYBRIDJOIN, *NUCES - FAST*

- Designed and implemented a near-real-time Data Warehouse prototype for Walmart using customer, product, and transactional datasets, applying a star schema for multidimensional analysis (slicing, dicing, drill-down, roll-up).
- Implemented the HYBRIDJOIN algorithm in Python to join fast-arriving transaction streams with master data, using hash tables, queues, disk buffers, and stream buffers for efficient near-real-time processing.
- Gained hands-on experience in stream processing, ETL workflows, large-scale data handling, and business analytics, demonstrating the application of database theory to complex retail scenarios